Mohammad H. Seyedsalehi

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Overview

I am an economist specializing in empirical industrial organization, environmental and energy economics, and econometrics. I study pricing and competition in vertically linked markets, carbon policy design, dynamic firm behavior, and climate finance. My econometric research focuses on empirical Bayes methods and the structural estimation of dynamic models.

Education

University of Toronto, Toronto, Canada. Expected June 2027
PhD in Economics
Fields: Industrial Organization, Environmental and Energy Economics, Econometrics
Committee: Victor Aguirregabiria (co-supervisor), Stephan Heblich (co-supervisor), Robert McMillan, Eduardo Souza-Rodrigues

Sharif University of Technology, Tehran, Iran. 2021
MSc in Economics (Finance)

Sharif University of Technology, Tehran, Iran. 2018
BSc in Computer Engineering

Working Papers

1. **Decarbonization Policy Along the Fuel Supply Chain: A Structural Analysis of Brazil's RenovaBio [Link]**
Solo-authored; *Job Market Paper*
Presentations: Toronto Meeting on the Economics of Climate Change (2026); University of Toronto, Department of Economics (IO Seminar, Climate Change Seminar); Rotman School of Management (EAP/IP Seminar)

Papers Under Review

1. **Empirical Bayes with Side Information: Incorporating Geographic Relevance into Neighborhood Impact Estimation [Link]**
Solo-authored; *Journal of Applied Econometrics (Second-round revise and resubmit)*

Publications

1. **The Effects of Order Flow Imbalance on Stock Prices in Tehran Stock Exchange [Link]**
with Mahdi Barakchian; *Journal of Financial Management Perspective, 12(40), 9–27, 2022 (in Persian).*

Work in Progress

1. **Euler Equations for Dynamic Games**
with Victor Aguirregabiria

2. **Who Benefits from Green Subsidies? Labor Standards and the Distribution of Gains**
   with Zahra Sedaghat Telgard
3. **One Carbon Market or Many? Optimal Integration of Sectoral Carbon Markets**

Honors, Awards and Fellowships

University of Toronto: Doctoral Fellowship ($20,000 × 5); Tuition Fellowship ($9,000 × 5); Doctoral Recruitment Award ($10,000); Doctoral Completion Award	2021–2026
Exceptional Talent Scholarship for MSc, Sharif University of Technology	2018–2020
Exceptional Talent Scholarship for BSc, Sharif University of Technology	2014–2018
Silver Medal, 22nd International Mathematics Competition (IMC), Bulgaria	2015
Silver Medal, 55th International Mathematical Olympiad (IMO), South Africa	2014
Bronze Medal, 54th International Mathematical Olympiad (IMO), Colombia	2013

Teaching

University of Toronto — *Teaching Assistant* September 2021–Present

Machine Learning Applications in Economics and Finance (7 semesters)	Undergraduate/Graduate
Financial Economics, Banking, and Corporate Finance (4 semesters)	Undergraduate
Financial Risk Management (3 semesters)	Undergraduate/Graduate
Special Topics in Economics with Data Analytics	Undergraduate
Microeconomics and Microeconomic Theory	Undergraduate/Graduate
Macroeconomics and Global Macroeconomic Policy	Undergraduate
International Trade and Monetary Economics	Undergraduate
Climate Economics	Undergraduate/Graduate
Economic Analysis of Law	Undergraduate
Classical Economic Thought	Undergraduate

Employment

University of Toronto — *Research Assistant* Summer 2023
To Professor Robert McMillan

Tosan Ofogh Investment Co., Tehran, Iran. Aug 2018–Aug 2021
Co-founded a startup within a hedge fund, conducting market research in the energy sector.

XaaS Cloud Computing Co., Tehran, Iran. Summer 2017
Software Engineering Intern

Big Data Lab, Sharif University, Tehran, Iran. Summer 2016
Research Intern

Skills

Python, R, MATLAB, Stata, SQL; machine learning; web scraping; Linux; Git

Other Information

Citizenship: Iranian
Languages: Persian (native), English (fluent)

References

Dr. Victor Aguirregabiria, Professor
University of Toronto, Department of Economics
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Dr. Rob McMillan, Associate Professor
University of Toronto, Department of Economics
Email: robert.mcmillan@utoronto.ca

Dr. Stephan Heblich, Professor
University of Toronto, Department of Economics
Email: stephan.heblich@utoronto.ca

Dr. Eduardo Souza-Rodrigues, Professor
University of Toronto, Department of Economics
Email: e.souzarodrigues@utoronto.ca

Abstracts

Decarbonization Policy Along the Fuel Supply Chain: A Structural Analysis of Brazil's RenovaBio

Solo-authored

Abstract. Incentivizing biofuel production is a widely used strategy for decarbonizing transportation, yet carbon emissions may rise if the additional biofuel fails to displace enough fossil fuel to offset its own emissions. Central to the outcome is the transmission of production incentives along the fuel supply chain, lowering cleaner-fuel prices in wholesale and retail markets and shaping both fuel substitution and overall demand. To trace these market adjustments, I develop and estimate a structural equilibrium model linking ethanol production, wholesale ethanol markets, oligopolistic fuel retailing, consumer demand, and a dynamic market that provides incentives for cleaner-fuel production through bankable credits. I apply the model to Brazil's RenovaBio program, under which biofuel producers earn credits that fossil-fuel distributors are required to purchase, financing the transition to cleaner fuels within the industry. The estimates imply that most of the production incentive reaches consumers through lower ethanol prices. Relative to a counterfactual without RenovaBio, the program increases ethanol production but leaves gasoline consumption almost unchanged, raising total fuel energy consumption by approximately 0.7% over 2020–2024. Despite official government reports of emissions savings equivalent to roughly 15% of Brazil's transport emissions, I find that the program instead increases CO_2 emissions by 0.22%. Further, policy simulations indicate that the same ethanol production, along with 1.20% lower emissions and higher welfare, could be achieved by replacing RenovaBio with an ethanol blending requirement around one percentage point higher. The findings show that the environmental effects of biofuel incentives depend crucially on market adjustments throughout the fuel supply chain, rather than on cleaner production alone.

Empirical Bayes with Side Information: Incorporating Geographic Relevance into Neighborhood Impact Estimation

Solo-authored

Abstract. This paper introduces a novel methodology for estimating neighborhood impacts on intergenerational mobility. The approach employs a nonparametric empirical Bayes framework that addresses geographic heterogeneity in the underlying data-generating process. First, I propose a theoretical framework to address the long-standing yet unresolved question of how to make better use of highly relevant samples—in this context, data from geographically closer regions—when designing a shrinkage estimator for a target area. I demonstrate, both theoretically and empirically, how neglecting this factor can lead to systematically distorted estimates. Second, using publicly available data, I demonstrate that the relevance-based empirical Bayes method reduces Mean Squared Error (MSE) by 26% to 34% relative to benchmark estimates.

The Effects of Order Flow Imbalance on Stock Prices in Tehran Stock Exchange

with Mahdi Barakchian

Abstract. We investigate the effects of order-book events on the prices of the 30 largest stocks on the Tehran Stock Exchange during 2020. We measure the sensitivity of prices to changes in supply and demand volumes and identify the factors affecting this sensitivity. Following the approach of Cont et al. (2014), we conduct approximately 30,000 OLS regressions and show that, in a low-depth market, mid-price returns are significantly explained by order flow imbalance, which captures the net change in demand—that is, the difference between event volumes on the two sides of the order book. We also show that incorporating order flow imbalance from the first three levels of the order book increases the model's explanatory power for changes in stock mid-prices, indicating that deeper levels of the order book also significantly affect price changes. Moreover, the results confirm that market depth reduces the price impact of order-book events. Our findings are robust across months and stocks.

Euler Equations for Dynamic Games
with Victor Aguirregabiria

Abstract. Decisions about market entry, capital investment, and innovation are inherently forward-looking, requiring firms to anticipate both future market conditions and their competitors' reactions. This project develops a computationally tractable framework for recovering the dynamic incentives underlying these decisions in oligopoly markets. Rather than relying on repeated solutions of the dynamic game, we use estimated firm strategies to construct optimality restrictions that connect marginal payoffs across three periods. Exploiting finite-dependence properties, we derive Euler equations that remove continuation-value derivatives while preserving the effects of strategic interactions. The approach extends conventional Euler-equation methods to dynamic games and provides a practical foundation for structural estimation in models of entry and exit, investment, and innovation.

Who Benefits from Green Subsidies? Labor Standards and the Distribution of Gains
with Zahra Sedaghat Telgard

Abstract. How are the gains from green investment subsidies distributed between firms and workers? This project examines whether labor conditions attached to these subsidies increase workers' share of the benefits, using Canada's Clean Technology Investment Tax Credit. The policy offers a larger credit to firms meeting prevailing-wage and apprenticeship requirements, linking investment support to employment practices. The analysis links firms' investment and contractor choices to worker earnings. Firms can qualify for additional support without raising existing workers' pay if they already meet the standards or hire contractors that do. Higher wages at subsidized projects therefore need not reflect gains caused by the policy. Comparing subsidies with and without labor conditions at the same fiscal cost would assess whether these requirements increase worker earnings and training opportunities, and how they affect investment and hiring.

One Carbon Market or Many? Optimal Integration of Sectoral Carbon Markets

Abstract. This paper asks whether carbon credit markets that jointly penalize dirty activity and reward clean activity should be organized separately by sector or integrated into a common market. Integration can establish a common carbon price, equalizing marginal abatement costs across sectors and shifting emissions reductions toward lower-cost opportunities, while also allowing payments from dirty activity in one sector to finance clean investment in another. Separation, however, preserves sector-specific certificate prices, allowing clean incentives to respond independently to differences in innovation and adoption frictions. I study a market design in which clean activities generate tradable certificates and dirty activities create demand for the same asset, with sector-specific crediting and obligation rates determining the effective clean subsidy and carbon penalty. I characterize when a unified market can achieve the efficient combination of a common carbon price and differentiated clean-technology support, when separate sectoral markets perform better, and how this tradeoff depends on abatement costs and clean-technology spillovers across sectors.